LOCAL EFFECTS

Most coasts get two high tides a day. But some get just one, because the shape of the coastline alters the way the water flows. This also affects the height of the tide. On some shores, water forced into funnel-shaped bays causes very high tides.

TIDAL RACES

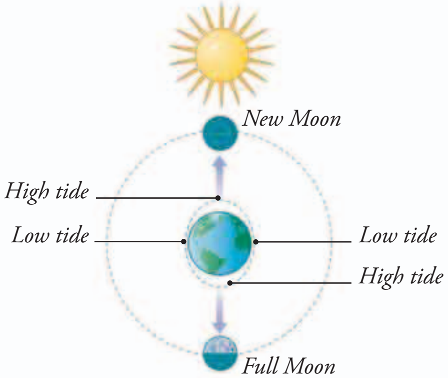
As the tide rises and falls, water is moved along coasts in local currents called tidal streams. Where these are forced around headlands and between islands, the flow speeds up, and sometimes causes dangerous tidal races and whirlpools. These only appear when the tidal stream is flowing fast, halfway between high and low tide. At other times, the water can be completely calm.

◀ MAELSTROM
The Maelstrom of Saltstraumen, on the northeast coast of Norway, is one of the world's most famous and dangerous tidal races. Twice a day, water surges through the strait at speeds of up to 25 mph (40 kph).

MOON AND SUN

Every two weeks, at full Moon and new Moon, the orbiting Moon falls in line with the Sun. At these times, the gravity of the Sun and Moon combine to cause extra-large tides called spring tides, with a big difference between high and low water level. At half Moon, the gravity of the Sun offsets the gravity of the Moon, causing neap tides with a much smaller difference between high and low water.

▶ SPRING TIDES
The tidal bulges are bigger when the gravity of the Sun and Moon are combined. This causes spring tides.



▶ NEAP TIDES
When the gravity of the distant Sun acts against the Moon's gravity, this makes the tidal bulges smaller.